Secure By Design

EDUCATE. EMPOWER. SECURE THE FUTURE.

COURSE 1.2 / COMPANION WORKBOOK

Configure DNS and time

Zones, records, the domain time hierarchy & troubleshooting

Build dependable internal name resolution and time on DC01 and DC02. Configure forward and reverse zones, manage service records, point the domain at the internal reference clock, and prove both in the air-gapped training network.

MATERIALS EDITION 1.2 / COURSE 1.2

Updated 8 October 2026 · sbd-dns-configuration

NAME _____ / DATE _____

© 2026 Secure By Design Team LLC.

Table of contents

Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.

Before you start

01 Map names to the network

02 Configure forward and reverse zones

03 Create and maintain service records

04 Prove resolution and directory discovery

05 Stand up SBD-TIME01 and the time hierarchy

06 Find the fault and record the fix

Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

Before you start

The addresses and names below continue the training example. Match every change to your own address plan. This network has no internet DNS or time dependency: names resolve on the domain controllers, and time comes from the internal reference clock, SBD-TIME01.

Prerequisites

- ☐ Complete the DC01 + DC02 identity build with both domain controllers healthy.
- ☐ Use an isolated Windows Server 2025 practice environment and an account delegated to manage DNS.
- ☐ Continue the existing sbd.test example: SBD-DC01 is 10.0.10.10 and SBD-DC02 is 10.0.10.11 on 10.0.10.0/24.
- ☐ The network's routing and VLAN segmentation are already in place, with a documented service-path register; SBD-TIME01 answers at 10.0.20.40 (a stand-in Windows Server 2025 host is fine in a practice lab).

What this course builds

- Plan authoritative internal DNS and client settings.
- Configure secure AD-integrated zones and maintain records.
- Test both resolvers, directory discovery, and client fallback.
- Point the PDC emulator at SBD-TIME01 and verify the domain time hierarchy end to end.

Systems in scope

RECORD	YOUR OBSERVATION
SBD-DC01 — name, address, OS build	
SBD-DC02 — name, address, OS build	
SBD-TIME01 — name, address, OS build	

© 2026 Secure By Design Team LLC.

LESSON 01 / ABOUT 15 MIN

Map names to the network

WHY THIS MATTERS A DNS plan explains which server owns a name and how a client finds that server. Begin with the working directory build, then extend it deliberately.

In this lesson you will

- Explain which DNS servers a domain member must use in this lab and why.
- Inventory the existing sbd.test and _msdcs zones without creating duplicates.
- Document the forwarding and recursion decision for an air-gapped network.

Follow the build

1. Record the domain, each subnet, both DNS addresses, and the owner of each service name. Keep host records and service aliases distinguishable.
2. On each domain controller, open DNS Manager and review the existing sbd.test and _msdcs zones. Promotion may already have created them; do not create duplicate zones.
3. Check the DNS role and static network settings. Domain members must use internal DNS servers that resolve the AD namespace; a public resolver is not an alternate for this environment.
4. Decide whether any separate internal namespace needs a conditional forwarder. With no recursive dependencies, document a no-recursion design; do not add internet forwarders or delete every root hint as a shortcut.

Check your result

```
Get-WindowsFeature DNS
Get-DnsClientServerAddress -AddressFamily IPv4
Get-DnsServerZone -ComputerName SBD-DC01
```

WHAT SUCCESS LOOKS LIKE Both servers and the client point to the intended internal DNS service. Existing AD zones and the forwarding/recursion decision are recorded.

Your record

Evidence to keep: Save the address/name worksheet and the initial zone inventory before changing anything.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	

© 2026 Secure By Design Team LLC.

RECORD	YOUR OBSERVATION
Open items or findings	

Think it through

Which DNS servers should a domain member use in this lab?

> **REMEMBER** Domain members point only at the internal DNS servers that resolve sbd.test. The existing AD zones are inventoried, not recreated. The forwarding and recursion decision is written down before anything changes.

Microsoft · Install and configure DNS Server

LESSON 02 / ABOUT 20 MIN

Configure forward and reverse zones

WHY THIS MATTERS Forward lookup answers where a name lives. Reverse lookup answers which name is assigned to an address. AD integration lets the directory replicate zone data between the selected DNS servers.

In this lesson you will

- Confirm sbd.test is AD-integrated with the intended replication scope.
- Create a secure, AD-integrated reverse zone for 10.0.10.0/24.
- Explain how zone data moves between the two DNS hosts.

Follow the build

1. In DNS Manager, inspect sbd.test Properties. Confirm that it is AD-integrated and choose the domain or forest replication scope that matches the DNS servers intended to host it.
2. Use secure dynamic updates for the AD-integrated zone. Review delegation and permissions before broadening who can change records.
3. Under Reverse Lookup Zones, create an IPv4 primary AD-integrated zone for network ID 10.0.10 if an equivalent zone does not already exist. Select the approved replication scope and secure updates.
4. Review zone-transfer settings. AD replication between these DCs does not require enabling transfers to arbitrary DNS servers; restrict any separately required secondary-server transfers explicitly.

Check your result

```
Get-DnsServerZone -ComputerName SBD-DC01 |
  Select-Object ZoneName, ZoneType, IsDsIntegrated, DynamicUpdate, ReplicationScope
Get-DnsServerZone -ComputerName SBD-DC02
```

WHAT SUCCESS LOOKS LIKE The intended forward and reverse zones appear on both DNS hosts after replication. The reverse zone for this /24 is 10.0.10.in-addr.arpa.

Your record

Evidence to keep: Capture the zone type, update policy, replication scope, and transfer policy for each zone.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	

© 2026 Secure By Design Team LLC.

RECORD	YOUR OBSERVATION
Open items or findings	

Think it through

How does an AD-integrated zone reach the other DNS-enabled domain controller in its replication scope?

> **REMEMBER** Forward and reverse zones are AD-integrated with a deliberate replication scope. Dynamic updates are secure-only. Zone transfers stay restricted; directory replication moves the data.

Microsoft · Manage DNS zones

LESSON 03 / ABOUT 20 MIN

Create and maintain service records

WHY THIS MATTERS Build a small, intentional record set. A host record identifies a machine; an alias provides a stable service name without making that alias a separate server.

In this lesson you will

- Distinguish host records (A and PTR) from service aliases (CNAME).
- Create a host record with its PTR and a service alias that match the plan.
- Maintain records with recorded ownership, lifetimes, and reasons.

Follow the build

1. Review the DC A and PTR records first. Correct mismatches through the owning host and DNS administration process rather than blindly adding another record with the same name.
2. Choose one approved service host from your address plan. In sbd.test, use New Host (A or AAAA), enter its actual name and address, and select the associated PTR option when the reverse zone is ready.
3. For a service alias, use New Alias (CNAME) and select the intended fully qualified host. For the later PKI course, reserve pki.sbd.test for the approved internal publication host; do not point it at the offline root.
4. Set the record lifetime to the operational plan, record static ownership, and remove only records confirmed obsolete. An alias cannot coexist with other record types at the same name.

Check your result

```
Get-DnsServerResourceRecord -ComputerName SBD-DC01 -ZoneName sbd.test
Resolve-DnsName SBD-DC01.sbd.test -Type A -Server 10.0.10.10
Resolve-DnsName 10.0.10.10 -Type PTR -Server 10.0.10.10
```

WHAT SUCCESS LOOKS LIKE Forward and reverse responses agree with the address plan. Each alias reaches the intended host; creating a DNS name alone does not configure its service.

Your record

Evidence to keep: Keep a record table with owner, type, target, time to live, and reason for the change.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	

© 2026 Secure By Design Team LLC.

RECORD	YOUR OBSERVATION
Open items or findings	

Think it through

What does creating pki.sbd.test as an alias accomplish?

> **REMEMBER** Host records identify machines; aliases provide stable service names. pki.sbd.test is reserved for the internal publication host, never the offline root. Every record carries an owner, a lifetime, and a reason.

Microsoft · Manage DNS resource records

LESSON 04 / ABOUT 20 MIN

Prove resolution and directory discovery

WHY THIS MATTERS Test each server explicitly before testing client fallback. A successful cached answer does not prove that both DNS servers hold the right records.

In this lesson you will

- Query each DNS server explicitly instead of trusting cached answers.
- Verify the LDAP service records that let clients discover domain controllers.
- Run a controlled failover test and record the result.

Follow the build

1. From a domain client, query each DNS server for the same host and reverse record. Record the server queried and the returned data.
2. Query the AD LDAP service records below against both servers. These records let a client discover domain controllers; an ordinary host lookup does not prove that discovery works.
3. Run the DNS diagnostics and directory replication checks on a domain controller. Investigate failures in context, including the intentionally isolated network design.
4. During an approved lab test, make one DNS server unavailable to the test client and verify resolution through the other. Restore connectivity and repeat for the opposite server. Do not interrupt shared production services for this exercise.

Check your result

```
Resolve-DnsName SBD-DC01.sbd.test -Server 10.0.10.10
Resolve-DnsName SBD-DC01.sbd.test -Server 10.0.10.11
Resolve-DnsName _ldap._tcp.dc._msdcs.sbd.test -Type SRV -Server 10.0.10.10
Resolve-DnsName _ldap._tcp.dc._msdcs.sbd.test -Type SRV -Server 10.0.10.11
dcdiag /test:dns /v
repadmin /replsummary
```

WHAT SUCCESS LOOKS LIKE Both DNS servers return the expected host and AD service records, replication is healthy, and the client still resolves internal names when either resolver is unavailable.

Your record

Evidence to keep: Keep paired lookup results, diagnostic output, and the failover test record with start, restore, and final verification times.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

Why query each DNS server directly?

REMEMBER Both resolvers answer the host and SRV queries correctly. dcdiag and repadmin confirm DNS and replication health. The failover test is documented with times, not just a passing screenshot.

Microsoft · Resolve-DnsName

Microsoft · Dcdiag

LESSON 05 / ABOUT 35 MIN

Stand up SBD-TIME01 and the time hierarchy

> **WHY THIS MATTERS** Kerberos tolerates minutes of skew, not more, and an air-gapped network has no internet time to lean on. Point the PDC emulator at the internal reference clock, SBD-TIME01, and let the domain hierarchy carry time to everything else.

In this lesson you will

- Confirm SBD-TIME01 is serving as the network's internal reference clock.
- Point the PDC emulator at it and keep the rest on the domain hierarchy.
- Document the accuracy, fallback, and drift decisions the model calls for.

Follow the build

1. Confirm SBD-TIME01 at 10.0.20.40 answers as an NTP server. In the model it is a rack appliance; in a practice lab a small Windows Server 2025 host with the Windows Time service configured as an NTP server can stand in for it. In an air-gapped network its clock is the reference by declaration — record the source decision (an approved hardware source, or the stand-in host's clock) with its expected accuracy and drift threshold.
2. On SBD-DC01, the domain controller holding the PDC emulator role, configure Windows Time to use SBD-TIME01 as its manual peer and mark it reliable. Only the PDC emulator points at TIME01: SBD-DC02 syncs from the PDC emulator, every other domain member keeps the default domain-hierarchy sync from the domain controllers, and the domain-joined Linux servers in later courses point chrony or ntpd at both domain controllers.
3. Confirm the network's service-path register permits time — members to both domain controllers on UDP 123, and SBD-DC01 alone to SBD-TIME01. Write the time design into the build record now: course 1.3's hardening Group Policy objects (GPOs) must leave it intact, and if a Security Technical Implementation Guide (STIG) value disagrees with it, that course's deviation record is where the decision lives.
4. Verify the chain end to end: TIME01 serving, the PDC emulator sourcing from it, SBD-DC02 and a member server sourcing from a domain controller, and offsets within your recorded threshold.

Check your result

```
Get-ADDomain | Select-Object PDCEmulator   # expected: SBD-DC01.sbd.test
# On the PDC emulator, then on SBD-DC02 and a member server:
w32tm /query /source
w32tm /query /status
w32tm /stripchart /computer:10.0.20.40 /samples:3 /dataonly   # from SBD-DC01 only
```

> **WHAT SUCCESS LOOKS LIKE** The PDC emulator reports SBD-TIME01 as its source, SBD-DC02 and the member server report a domain controller, and measured offsets sit inside the documented threshold.

Your record

Evidence to keep: Record the time design decisions, the source and status output from both controllers and the member, the stripchart offsets, and the register rows that permit time: members to the domain controllers, and SBD-DC01 to SBD-TIME01.

© 2026 Secure By Design Team LLC.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Which system syncs directly with SBD-TIME01?

> **REMEMBER** TIME01 is the declared reference; its accuracy decisions are written down. Only the PDC emulator peers with it; the hierarchy carries the rest. Offsets are measured and inside the threshold, not assumed.

Microsoft · Windows Time service

Microsoft · Windows Time service tools and settings

LESSON 06 / ABOUT 15 MIN

Find the fault and record the fix

WHY THIS MATTERS Troubleshoot one layer at a time: client settings, server reachability, authoritative data, replication, and finally the application that uses the name.

In this lesson you will

- Isolate a DNS failure one layer at a time.
- Distinguish a missing name (NXDOMAIN) from timeouts and service failures.
- Record the cause, the fix, and the rollback — not just a passing result.

Follow the build

1. For a failing lookup, record the exact name, record type, client, resolver, timestamp, and error. Compare the fully qualified name with any short-name search behavior.
2. Query both servers directly. If they disagree, inspect zone scope and replication. If both are correct but the client is not, inspect client DNS settings and cached data.
3. Distinguish a missing name (NXDOMAIN) from a timeout or service failure. Fix the owning configuration first, then use Clear-DnsClientCache on the test client if stale cache remains relevant.
4. Repeat the original test and the redundancy checks. Record what changed and the rollback method. Review aging/scavenging separately against DHCP, static records, and backup practices before enabling automatic deletion.

Check your result

```
Get-DnsClientServerAddress -AddressFamily IPv4
Get-DnsClientCache
Resolve-DnsName SBD-DC02.sbd.test -Server 10.0.10.10
Resolve-DnsName SBD-DC02.sbd.test -Server 10.0.10.11
```

WHAT SUCCESS LOOKS LIKE The original failure is reproducibly resolved, both resolvers agree, and the change record explains the cause rather than only showing a passing screenshot.

Your record

Evidence to keep: Complete the DNS checklist and retain the before/after lookup results with the configuration owner.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

© 2026 Secure By Design Team LLC.

RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

Both DNS servers return the correct address, but one client keeps using the old address. What should you inspect next?

> **REMEMBER** Troubleshoot client settings, reachability, authoritative data, replication, then the application. NXDOMAIN, timeouts, and service failures point at different owners. The change record explains the cause and the rollback.

Microsoft · Manage DNS resource records

Microsoft · Resolve-DnsName

Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

Lesson completion

LESSON	EVIDENCE FILED	DATE
01 Map names to the network	☐ yes	
02 Configure forward and reverse zones	☐ yes	
03 Create and maintain service records	☐ yes	
04 Prove resolution and directory discovery	☐ yes	
05 Stand up SBD-TIME01 and the time hierarchy	☐ yes	
06 Find the fault and record the fix	☐ yes	

Open items carried forward

Next in the learning path

That completes Configure DNS and time. Course 1.3, Harden the baseline, builds the organizational units and hardened security policies that every later system joins.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

© 2026 Secure By Design Team LLC.